

statements that can be executed on command. The use of these scripts can eliminate the transmission time from the PC to the switch system because all commands and statements in the script are executed by the mainframe. For switch mainframes equipped with an LXI interface, all TSP configuration and script loading and data transfer can be performed via this interface," explains Leutbecher.

What's coming

If the last decade is anything to go by, the next one holds tremendous promise for the test and measurement industry.

"We will see ultra-high-speed networks making it possible to access all the content from the Web online. This could make test and measurement platform-independent and available through the provider's portal as a service harnessing the power of multiple connected systems to perform complex measurements and analysis. Perhaps, a more modular app like system would be the order of the day with minimal footprint on the client device

Key challenges for manufacturers

1. Lack of electronic devices, discrete or embedded, indigenously available
2. Hybrid equipment thwarting the sales of discrete testers
3. Constantly evolving standards and technology, particularly in wireless
4. Growing rental market restraining the sale volumes
5. Heavy influx of imports, especially low-value products
6. Misperception about the use of T&M instruments
7. Lack of qualified, trained hardware and RF engineers
8. Price-sensitive customers and hence price constraints
9. High initial investments
10. Trading more profitable than manufacture under present policies
11. In-house demand volumes are very small
12. Lack of motivation and favourable policies from the government

—Neelam K. Kumar, Executive Director,
Aplab Limited, India

and demand-based selection of features online," says Yasir Fahim, general manager, ADInstruments.

New human interface devices like touch, gesture and speech recognition will take centre stage, making it possible to analyse data like never before. "Touch-screen panels and ultra-portability are expected to translate into significant leaps forward for the test and measurement industry," says Mohammed Ghouse, country manager (business communication), Scientech Technologies.

Wireless sensing will get further refined and more accurate. Technologies such as wireless charging for telemetrical probes will establish in the mainstream right down to the consumer electronics level.

According to Mrs Kumar, "Demand for synthetic instrumentation (with high performance, less footprint, ruggedness, high levels of integration, faster testing, flexibility and reconfigurability) is probably the next big wave in T&M."

Ultra-fast I-V sourcing and measurement capabilities are becoming increasingly critical for many technologies, including compound semiconductors, medium-power devices, non-volatile memories, micro-electromechanical devices (MEMS), nano devices, solar cells and CMOS devices as well as high-power discrete or so-called high-brightness LEDs. Using pulsed I-V signals to characterise devices rather than DC signals makes it possible to study or reduce the effects of self-heating (joule heating) or to minimise current drifting in measurements due to trapped charge.

Transient I-V measurements allow scientists and engineers to capture ultra-high-speed current or voltage waveforms in the time domain or study dynamic test circuits. Pulsed



Caddo iSeries by Scientech Technologies

sourcing can be used to stress test a device using an AC signal during reliability cycling or in a multi-level waveform mode to program/erase memory devices.

Emerging 3G and 4G communications standards are driving investment in the performance products and mainstream embedded applications are increasing in demand.

Spectrum analysers are also seeing an increase in demand as needs for greater wideband performance in radar and spectrum management increase, particularly in intelligence, regulatory monitoring and defence applications.

Manufacturers face challenges, though

The challenges before T&M manufacturers can be summarised in three words—smaller, faster and cheaper. In other words, their products must be able to characterise ever-smaller components more quickly and less expensively. Nowhere is this more apparent than in the development of instrumentation for nanotechnology research.

Nanotechnology has the potential to improve our quality of life in diverse ways, such as faster electronics, huge memory/storage capacities for PCs, cheaper energy through more efficient energy conversion, and improved security through the development of nanoscale bio- and chemical-detection systems. However, before these new technologies become commercial realities, researchers must be able to characterise nano material and device